

Data and Code Management in 2026

From Collection to Application

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Block 1 — Why DCM now?

Connect with your previous courses	5 min
What AI changes — and what remains essential	6 min
Junior data work: evidence and discussion	9 min
From analysis to reliable, usable work	5 min
Our course map: four stages, AI throughout	20 min
Teaching block	45 min

Why does DCM matter when AI can increasingly write code?

You already know how to analyse

Previous courses introduced coding, statistics, analytics, BI or machine learning. You have used these to answer questions and communicate results.

DCM asks the next question

Can another person **understand, check, reproduce and use** your work?

AI can help produce code. You still need to understand the problem and data, judge the solution, and provide evidence that it is correct.

AI changes where effort goes

You have practised

Writing code

Learning syntax

Producing an answer

Working in a script or notebook

Completing an analysis

DCM adds emphasis on

Specifying, inspecting and modifying code

Reasoning about data, logic and systems

Testing the result and its assumptions

Sharing a reproducible project

Delivering something others can use

Code production becomes more AI-assisted. Code literacy stays essential.

What this means for junior data professionals

- **WEF employer survey:** analytical thinking, technological literacy and AI/data skills remain important or are expected to grow. [1]
- **OECD evidence:** AI changes tasks and skill requirements; effects vary across occupations and workplaces. [2]

Our practical advice

Be able to understand a request, inspect an implementation, validate its output and collaborate on a reliable result.

These sources do not establish that AI is replacing junior roles. References and limitations are at the end.

Discussion — where does AI change your work?

Think of a recent programming, statistics, ML or BI assignment.

1. Which part would you ask AI to help with?
2. Which part would you **never accept without checking**?
3. What evidence would make you trust the result?

Two minutes in pairs; then two examples from the room.

From analysis to production

Knowing how to analyse

- choose a method;
- explore or model data;
- produce a result.

Producing usable data work

- trace data and decisions;
- check results and review changes;
- rerun in a known environment;
- deliver something others can use.

Reliable + reproducible + usable

The course map: four stages, AI throughout



Read and reason → understand data → build reliably → make it usable.

AI assistance grows as your ability to supervise it grows.

1. Code literacy 2. Data

Read, reason, debug

- Python data structures and tables;
- control flow, functions, inputs/outputs;
- inspect, modify and debug generated code.

Start with projects, reproducibility and Git; then two Python literacy sessions.

Know what the data mean

- relational data, SQL, joins and aggregation;
- APIs, JSON and external/web data;
- provenance and validation of transformations.

Check keys, units, missing values and totals; getting data is only the start.

3. Engineering 4. Production

From scripts to maintainable projects

- functions/modules and project structure;
- documentation, tests and refactoring;
- Git/GitHub review and reproducibility.

More generated code makes testing, review and structure more important.

From analysis to something usable

- dependencies and environments;
- Docker and reproducible execution;
- Streamlit and a small data product.

Someone else should be able to run the work and use its results.

5. AI throughout — growing assistance, continuing judgment

Early: build literacy

Predict, write and explain short code.
Use AI to explain and help debug.

Later: supervise larger tasks

Generate, refactor, test, review and build
with AI assistance.

Understand → specify → inspect/plan → modify
→ inspect the diff → run → verify

You choose the checks and explain the result.

Block 2 — How this course will work

Semester roadmap	8 min
Learning outcomes	3 min
Two individual homeworks and one group project	8 min
AI expectations	4 min
Tool stack	4 min
Coding-agent preview	13 min
Setup, next week and recap	5 min
Teaching block	45 min

Roadmap I — foundations and responsible use

Opening

17 Sep: special workshops; no normal DCM class.

24 Sep: Introduction — DCM in 2026.

1. Code literacy

1 Oct: Reproducibility, projects, Git/GitHub.

8 Oct: Code literacy I.

15 Oct: Code literacy II.

5. AI throughout — dedicated guidance

22 Oct: asynchronous Responsible AI at UNIL.

Roadmap II — data to a usable product

2. Data

29 Oct: SQL & relational data.

5 Nov: APIs, JSON & web data.

3. Engineering

12 Nov: Software engineering with AI.

4. Production

19 Nov: Environments & Docker.

26 Nov: Applications + project kickoff.

Full integration — group project

3 Dec: Project work I.

10 Dec: Project work II/review.

17 Dec: Presentations.

AI throughout: explain, debug, generate, refactor, test, review, build.

What you should be able to demonstrate

Code literacy Read, explain, modify and debug Python in a reproducible project.

Data Work with files, databases and APIs; validate transformations.

Engineering Structure, document and test code; collaborate through Git.

Production Reproduce the environment and deliver a small usable application.

AI throughout: supervise assistance and defend the code, checks and choices.

Two individual homeworks + one group project

HW1 — individual

Code literacy + reproducibility

After the early block

Explain/debug code; tests; project/Git and rerun evidence.

HW2 — individual

Data + engineering + production

Later technical blocks

SQL, data validation, modular/testable code; environments and Docker where appropriate.

Project — group

Full integration: all four stages

26 Nov–17 Dec

A usable data product with checked results and defended choices.

AI-assisted work and individual understanding matter in all three.

Assessment — show the evidence behind the result

Explain	Code, data meaning, assumptions and limitations.
Validate	Checks, tests and independent evidence of correctness.
Reproduce	Inputs, instructions, dependencies and a fresh run.
Defend	Important choices, contributions and reviewed changes.
Account for AI	Appropriate assistance, verification and disclosure.

Individual understanding is required in both homework and group work.

Details to be confirmed

Grading percentages, homework deadlines and submission/group rules.

AI use is part of the course — responsibility is yours

AI assistance is permitted and encouraged **according to the task**.

- Understand and be able to explain anything you submit.
- Inspect generated code and independently verify outputs.
- Use appropriate tools and disclose assistance as instructed.
- Do not upload confidential or sensitive data to arbitrary services.

22 Oct: asynchronous Responsible AI at UNIL, with detailed guidance.

Today's demonstration uses only synthetic data.

Tools with a purpose

VS Code	Workspace: files, editor, terminal and source control.
Python	Default live language for data work and checks.
Git/GitHub	Change history, review and collaboration.
AI coding agent	Explain and implement scoped tasks under supervision.
Google Colab	Notebook environment and temporary setup fallback.
SQL	Query, join and aggregate relational data.
Docker — later	Reproducible execution environment.
Streamlit — later	A usable data application.

R remains available where appropriate, particularly for projects.

A preview of our workflow — inspect before delegating

Watch the instructor open the [Orders starter README](#) and start an agent **inside that project folder**.

First prompt

Inspect this project. **Do not modify anything yet.** Explain its structure, what it does, and how to run it.

Check the explanation against the files; run the baseline:

```
python summary.py  
python check.py
```

Expected: 4 synthetic orders; revenue CHF 50.00; checks pass.

A preview of our workflow — change, inspect, verify

One bounded request

In `summary.py` only, add average order value in CHF to two decimals. Preserve count and revenue. Do not change data/checks or add dependencies. Explain your plan before editing.

```
git diff -- summary.py
python summary.py
python check.py --expect-average
```

Inspect: only the intended change; no altered data or weakened checks.

Verify independently: $(12 + 18 + 7 + 13)/4 = 12.50$ CHF per order.

Prepare for 1 October — reproducible projects and Git

Use the [Setup checklist](#) — **minimum-tool section only**:

- open your starter folder in VS Code;
- run `summary.py` and `check.py` with Python;
- check Git and access to your GitHub account;
- try a read-only inspection with your coding agent.

Local setup blocked? Use the [Fallback notebook](#) in Colab, record the error, and complete local setup afterwards. No paid AI product is required.

Next week uses Python and Markdown. Skip the guide's older Quarto/Jupyter requirement. Docker comes later.

From Collection to Application — what ties it together

Code literacy → Data → Engineering → Production

AI throughout

Assistance can grow. Your ability to understand, inspect and verify must grow with it.

HW1: foundations. **HW2:** later technical blocks.

Group project: integrate the whole workflow into something usable.

Be ready to **explain, validate, reproduce and defend** your work.

References for the job/skills discussion

1. World Economic Forum (2025), *The Future of Jobs Report 2025*, chapter 3, Skills outlook. Employer expectations for 2025–2030; evidence about skills, not a forecast for every role.

[WEF: Skills outlook](#)

2. OECD (2024), *How is AI changing the way workers perform their jobs and the skills they require?* Policy brief; changing task and skill requirements, with differences across occupations and workplaces.

[OECD policy brief, 29 November 2024](#)

Sources checked 17 September 2026. No adoption statistics or employment forecasts are claimed. Advice to junior professionals is our pedagogical interpretation; these sources do not establish a universal loss of junior jobs.